Master's Thesis

Task description for Mister Ahmed Galai

Course of studies: Mechatronik und Robotik
Matriculation number: 10007404

Title: Augmented Reality-Enhanced Programming by Demonstration: 6D Pose Estimation Using Apple Vision Pro for Intuitive Robot Teaching

Title (ger.): Augmented Reality-unterstützte Programmierung durch Demonstration: 6D-Posenbestimmung mittels Apple Vision Pro für intuitive Roboterprogrammierung

General information:

Modern industrial production confronts a fundamental challenge: achieving comprehensive automation while preserving the flexibility to adapt to evolving production requirements and market demands. Human-Robot Collaboration (HRC) presents a promising solution, delivering sustainable improvements in automation levels and productivity across manufacturing sectors. Within HRC, "Programming by Demonstration" (PbD) represents a particularly transformative approach that eliminates the need for specialized programming expertise in robotic system configuration. This methodology empowers skilled workers – regardless of educational background – to directly interact with and program robotic systems for new tasks. Augmented Reality (AR) technologies can significantly enhance this paradigm by providing intuitive user interfaces and real-time visual guidance seamlessly integrated into physical work environments. However, the industrial viability of AR-supported programming approaches requires a precise and robust 6D pose estimation – a technical requirement that has historically constrained widespread adoption. Recent advances in machine learning and artificial intelligence have yielded significant breakthroughs in methods for 6D pose estimation, delivering superior accuracy, speed, and robustness compared to traditional computer vision approaches. The convergence of these AI-driven algorithmic improvements with advanced consumer AR devices, exemplified by the Apple Vision Pro, signals a potential paradigm shift. This technological convergence may finally provide the foundation necessary to bridge the gap between accessible, user-friendly interfaces and the exacting precision requirements of industrial automation applications.

Task description:

The primary objective of this master's thesis is to develop, implement, and evaluate a proof-of-concept application that leverages the Apple Vision Pro's spatial computing capabilities for 6D pose estimation in human-robot collaboration scenarios. The developed application will serve as a foundation for future research and development efforts by implementing the core functionalities for 6D pose estimation and visualization. The recording and analysis of specific assembly sequences falls outside the scope of this work.

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The application must be developed using Apple's official SDKs and should provide a user-friendly interface with appropriate interaction modalities for selecting a specific Region of Interest through gaze detection or similar input methods. The application must establish connections to external backend services that provide 6D pose estimation methods through an API interface. The estimated 6D pose of selected objects should then be visualized by integrating the pose data with the Apple Vision Pro's spatial tracking system, ensuring stable virtual object positioning within the user's environment. To enable future system enhancements, the application should implement continuous streaming of sensor data to external services using appropriate communication protocols.
The feasibility of the system must be evaluated and optimization measures should be implemented as necessary to demonstrate the practical applicability of the developed solution for human-robot collaboration scenarios.

In the context of this work the following tasks arise in particular:

- Research previous approaches for PbD using AR/MR using 6D pose estimation
- Identification of limitations regarding the usability of existing solutions and systems
- Definition of requirements for a proof-of-concept application using the Apple Vision Pro
- Development and implementation of the application
    - User Interface with suitable modalities for selecting the region of interest
    - API-based communication with backend services
    - Visualization components for displaying 6D pose estimation results within the mixed reality environment, based on object 3D model and spatial tracking
- Testing and evaluation using a basic assembly task as a demonstration example
- Written documentation of the results
- Presentation of the thesis

The processing time depends on the examination regulations applicable to the student.

The work is carried out in close contact with the supervisor appointed by the examiner. Reports on the progress of the work shall be submitted at regular intervals. The thesis must be submitted to the Institute in two permanently bound copies (glue, thermal or book binding) and on a CD/DVD or USB stick no later than 6 months after the issue of the topic.

Supervisor: David Wendorff

match-Identifier: MA 06/25

Date of issue: 01.07.2025 Submission date: ______________

A Raatz
Prof. Dr.-Ing. Annika Raatz

01.07
Ahmed Galai